

Dust Particles and Gas+Dust Self-Gravity in the AthenaK Shearing Box: I. The Dust-Module Merge (Phase 4a)

athenak-multigrid tree, branch dust-multigrid

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Abstract

This document opens the *dust track* of the self-gravity program: Lagrangian dust particles in the shearing-box code with multigrid self-gravity on refined meshes, toward the gas+dust gravito-turbulent boxes of Baehr, Zhu & Yang (2022) with the radial pressure gradient they omitted. It records what the code does on test problems, phase by phase; the mechanisms are in the companion `dust_selfgravity_implementation.pdf`.

Phase 4a merges the sibling fork `athenak-dust` — particle-mesh dust with Benítez-Llambay–Krapp–Pessah implicit drag inside the Krapp et al. (2024) IMEX integrator, validated in its own repository — into the multigrid tree, and asks two questions: is the dust module unchanged by the merge, and is the gravity machinery unchanged by the dust module? Both answers are yes to every printed digit. On the merged tree the dust fork’s own acceptance numbers reproduce exactly: the N -species damping ladder converges at second order in Δt with the reference errors 1.3529×10^{-7} , 3.4627×10^{-8} , 8.7724×10^{-9} , 2.2075×10^{-9} and total momentum conserved to 10^{-16} ; the multi-species Nakagawa–Sekiya–Hayashi drift equilibrium holds to 10^{-16} over ten orbits; the damped dusty sound wave converges toward second order in resolution; the ion-neutral C-shock under `imex2+` is bit-identical to its baseline. The gravity solver’s documented statics reproduce exactly (slab sheet error 1.203×10^{-12} , Tomida & Stone contraction 0.125, boundary-annuli shearing wave 5.950×10^{-3} / 2.057×10^{-3}), bit-identically at 4 ranks. The one defect found — a multi-rank deadlock of the additive deposit exchange against the tree’s rank-packed messaging — is fixed and ranks 1, 2 and 4 agree to all digits. Finally the gas-only stratified gravito-turbulence box evolved with `rk2` and with `imex2+` (the integrator the dust module requires) agrees in every history column to between 2×10^{-6} and 5×10^{-3} at $t = 3 \Omega^{-1}$, the expected spread between two second-order integrators.

A **second merge** the same day brought in the dust fork’s latest seven commits. The whole battery reproduces on the result to every digit (the gas-only run bit for bit), and the fork’s new predictor–corrector coupling under plain `rk2` (`coupling = pc2`) is exercised for the first time here: second order in the damping ladder (orders 2.23, 2.10, 2.04) and in the dusty wave (1.85, 1.98, 2.00), NSH to round-off, rank-independent at 1, 2 and 4 ranks — the integrator freedom that lets dust run under the same `rk2` as every gravity result before it.

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1 Plan and scope

1.1 The program

The science target is the interplay of gas gravitational instability and dust concentration: 3D stratified, cooling, gravito-turbulent shearing boxes with superparticles (Baehr, Zhu & Yang 2022: $512^2 \times 256$, $\beta = 10$ cooling with an irradiation floor, $\gamma = 5/3$, $\text{St} = 0.1, 1, 10$, $Z = 0.01$, $Q_0 = 1.02$), extended by the radial pressure gradient that admits the streaming instability, and eventually adaptive zoom on the dust clumps — the configuration their limitations section names. The route (implementation document, §1.2): Phase 4a merges the dust module; 4b adds what the 3D shearing box needs from it (shear-remapped additive deposits, ideal-gas back-reaction, a relaxed particle timestep); 4c adds dust self-gravity through the existing Poisson solve; 4d builds the science configuration; Phase 5 takes particles onto refined meshes.

1.2 What Phase 4a must establish

1. **Dust fidelity.** The dust fork ships its validation in `../athenak_dust_tests/` (notebooks with printed reference values). On the merged tree, those references must reproduce: the damping ladder (second order in Δt , momentum to round-off), the NSH drift equilibrium (round-off over ten orbits), the damped dusty wave (spatial convergence), the random-placement and single-block layout checks, and the regression of the ion-neutral C-shock that shares the `imex2+` coefficient code.
2. **Gravity regression.** Three static multigrid tests whose numbers are documented in the multigrid validation document: the slab-open sheet test, the Tomida & Stone §4.1 contraction, and the boundary-annuli shearing wave including its zero-phase bit-identity with the periodic twin.
3. **Rank independence.** Dust and gravity at 1, 2 and 4 MPI ranks.
4. **Integrator.** The dust module’s IMEX coupling requires `imex2+`; all gravity validation so far used `rk2`. A gas-only stratified gravito-turbulence box under both integrators, compared column by column.
5. **The second merge** (§4). After catching up with the dust fork: the battery again, and the new PC2/hybrid couplings under `rk2`.

2 Test setup

2.1 Builds

Serial and MPI builds in the tree’s `build/` and `build-mpi/`, both with `-D PROBLEM=built_in_pgens -D Athena_ENABLE_FFT=ON`, the MPI one with `-D Athena_ENABLE_MPI=ON` (Open MPI via Homebrew). Everything below was first run on a scratch clone at the same commits and then rerun with the tree’s own binaries; the two sets of results are identical, including the gravito-turbulence histories bit for bit.

2.2 Inputs and commands

Dust inputs come from the dust-tests repository (paths relative to the directory holding both trees):

```

# damping ladder (tlim = 1, so dt = 1/ncycle)
for c in 0.4 0.2 0.1 0.05; do
  athena -i athenak_dust_tests/inputs/dust_damping.athinput particles/ppc=3 time/
  cfl_number=$c
done
# layout variants
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false
athena -i .../dust_damping.athinput particles/ppc=4 problem/lattice=false dust/deposit=
  ngp
athena -i .../dust_damping.athinput particles/ppc=3 meshblock/nx1=32 meshblock/nx2=32
# NSH equilibrium, dusty wave, C-shock regression
athena -i .../dust_nsh.athinput
athena -i .../dusty_wave.athinput mesh/nx1=$n meshblock/nx1=$((n/2))      # n = 32..256
athena -i inputs/ion-neutral/perp_cshock.athinput time/integrator=imex2+ time/nlim=500
# rank independence (drifting particles cross MeshBlock and rank boundaries)
mpirun -np $np athena -i .../dust_damping.athinput particles/ppc=3 problem/v0=0.5 dust/
  taus_3=0.5
# gravity statics
athena -i inputs/tests/mg_poisson_slab.athinput
athena -i inputs/tests/ts41_sin3.athinput
athena -i inputs/tests/mg_poisson_shear_bndry.athinput problem/profile=shwave problem/
  time0=0.37
# integrator twin (gas only, uniform grid)
athena -i inputs/shearing_box/gravito_turb_amr_test.athinput mesh_refinement/refinement=
  none \
  time/tlim=3.0 [time/integrator=imex2+ time/cfl_number=0.3]

```

Each dust problem generator writes a <basename>-errs.dat line per run; the gravity statics print their metrics to standard output. All result files are archived under `validation/run/dust4a/` (see its `README.txt`), which the figure scripts `figures/plot_dust4a_*.jl` read.

3 Results

3.1 Dust fidelity

Damping ladder (Krapp et al. 2024 §3.1). Uniform gas at rest coupled to three dust species, $t_s = (0.01, 0.1, 1.0)$, $\epsilon_s = 1/3$ each, one particle of every species at every cell center so the discrete update per cell *is* the integrator’s map of the ODE system. Errors are against a fine-step RK4 reference; the first run has $\Delta t > t_{s,1}$ (stiff regime).

	Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$ \Delta P_{\text{tot}} $
	1.235×10^{-2}	1.3529×10^{-7}	1.3711×10^{-7}	1.4641×10^{-7}	6.8939×10^{-7}	1.6×10^{-16}
	6.211×10^{-3}	3.4627×10^{-8}	3.5096×10^{-8}	3.7325×10^{-8}	1.7630×10^{-7}	7.5×10^{-17}
	3.115×10^{-3}	8.7724×10^{-9}	8.8920×10^{-9}	9.4213×10^{-9}	4.4631×10^{-8}	2.0×10^{-16}
	1.558×10^{-3}	2.2075×10^{-9}	2.2377×10^{-9}	2.3649×10^{-9}	1.1225×10^{-8}	1.7×10^{-16}
order of err_u		1.98, 1.99, 1.99				
reference (dust tests, 2026-07-14)		$\text{err}_u = 1.3529 \times 10^{-7}, 3.4627 \times 10^{-8}, 8.7724 \times 10^{-9}, 2.2075 \times 10^{-9}; \Delta P \sim 10^{-16}$				

Every reference digit reproduces (Fig. 1). The layout variants: random positions with TSC and with NGP deposit give $\text{err}_u = 5.29 \times 10^{-6}$ and 5.77×10^{-6} (the 10^{-5} -level departure from

the homogeneous ODE is physical: density fluctuations couple to pressure) with momentum to 3.8×10^{-16} ; the single-MeshBlock lattice run reproduces the four-block ladder row to all digits, which is the halo-exchange exactness check.

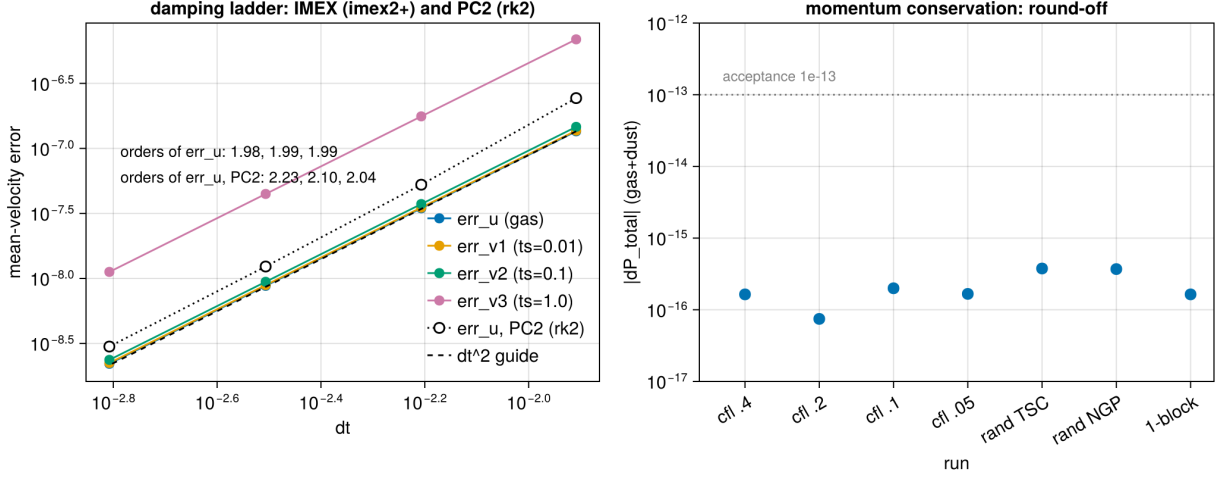


Figure 1: The damping ladder on the merged tree: second-order convergence of all four mean-velocity errors in Δt under the IMEX coupling (filled), the gas error under the PC2 coupling with **rk2** after the second merge (open, §4), and the total gas+dust momentum change of every IMEX run against the 10^{-13} acceptance line (right).

NSH drift equilibrium (2D r - z , ten orbits). Gas plus three species ($t_s = 0.01, 0.1, 1.0$; $\epsilon = 0.2, 0.5, 0.3$) started at the exact multi-species steady drift solution, with the gas forced by the pressure-gradient mimic $2\Omega\eta v_K$. After 4495 cycles every deviation is at round-off:

	$ \delta u_x $	$ \delta u_\phi $	$ \delta v_{x,1} $	$ \delta v_{\phi,1} $	$ \delta v_{x,2} $	$ \delta v_{\phi,2} $	$ \delta v_{x,3} $	$ \delta v_{\phi,3} $
serial	1.3×10^{-16}	6.2×10^{-17}	6.9×10^{-17}	5.9×10^{-17}	1.2×10^{-16}	2.4×10^{-17}	9.4×10^{-17}	4.3×10^{-17}
4 ranks	1.1×10^{-16}	9.0×10^{-17}	1.2×10^{-16}	2.8×10^{-17}	1.3×10^{-16}	1.0×10^{-17}	9.0×10^{-17}	3.3×10^{-17}

This is the test that certifies the rotation/forcing kicks and the implicit drag are composed unsplit inside the IMEX stages: any splitting error would show as secular drift.

Damped dusty sound wave (Laibe & Price 2011; Krapp et al. §3.2). The exact complex eigenmode of the linearized gas-dust system, L_1 errors of density and velocity against the analytic damped wave at $t = 2$:

N_x	cycles	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order
32	320	1.836×10^{-8}	1.351×10^{-8}	—
64	321	6.896×10^{-9}	5.124×10^{-9}	1.41
128	641	1.837×10^{-9}	1.377×10^{-9}	1.91
256	1281	4.765×10^{-10}	3.572×10^{-10}	1.95

The code's own root of the dispersion relation, $\omega = 4.529763 - 0.492158i$, is printed identically in every run. The order approaches two from below; the $32 \rightarrow 64$ step is polluted by the fixed-cfl

time step at the coarsest resolution (the same cycle count at 32 and 64 cells), as the dust-tests notebook also observes.

3.2 Regressions

The ion-neutral C-shock (`inputs/ion-neutral/perp_cshock.athinput`, `integrator = imex2+`, 500 cycles) is the direct consumer of the refactored `imex2+` coefficient code: $\text{RMS-}L_1 = 2.071216 \times 10^{-3}$, bit-identical to the dust fork’s baseline. The Sod tube runs through the untouched hydro task list.

3.3 Rank independence, and the deadlock

The damping problem with drifting particles ($v_0 = 0.5$, $t_{s,3} = 0.5$, three lattice species) sends particles across MeshBlock and rank boundaries many times per run and exercises per-stage migration, the additive deposit exchange, the u^* ghost fill and the back-reaction exchange.

The first attempt at 4 ranks **hung at cycle 0**. The cause was in the merge itself: the dust fork’s additive deposit exchange posted per-neighbor MPI messages against the base class of July’s upstream, while this tree carries upstream #775, whose base-class receive posting is rank-packed (one aggregated message per peer rank). The sends had no receiver. After moving the exchange onto the rank-packed path (implementation document §4.4) with the serial path untouched (the damping ladder is bit-identical before and after the edit):

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$ \Delta P_{\text{tot}} $
1	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	5.2×10^{-15}
2	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.7×10^{-15}
4	1.158578×10^{-5}	1.187422×10^{-5}	1.490167×10^{-5}	6.153323×10^{-5}	3.6×10^{-15}
reference	1.158578×10^{-5}	(dust tests: “1-rank and 4-rank identical”)			

The tiny spread in $|\Delta P|$ is the reduction order of the diagnostic itself. The NSH run at 4 ranks is in the table of §3.1.

3.4 Gravity regression

test	metric	measured	documented
<code>mg_poisson_slab</code> , sheet profile (absolute, no mean subtraction)	max rel / L_2 rel	1.203474×10^{-12} / 3.078736×10^{-13}	1.2×10^{-12}
<code>ts41_sin3</code> (Tomida & Stone §4.1)	defect ratio per V-cycle	0.1254, 0.1249, 0.1249, 0.1249	0.125
<code>mg_poisson_shear_bndry</code> , shwave, $t_0 = 0.37$	max rel / L_2 rel	5.950118×10^{-3} / 2.057024×10^{-3}	5.95×10^{-3} / 2.057×10^{-3}
same, $t_0 = 0$, shear-periodic vs periodic- x_1 twin	both runs	5.506611×10^{-3} / 1.281655×10^{-3} , identical	bit-identical

At 4 ranks the slab and boundary-annuli solves reproduce the serial digits exactly, as documented for Phases 2b and 3.

The dust module is inert without a `<dust>` block by construction (the hydro and gravity paths do not see it), and these numbers confirm the shared infrastructure it touched — driver coefficients, boundary-values base class, output tables, physics registration — left the solver untouched.

3.5 The integrator twin

The Phase-3 small stratified gravito-turbulence box (`inputs/shearing_box/gravito_turb_amr_test.athinput:16H×16H×12H`, $64\times64\times48$, $\beta = 10$, slab-open multigrid gravity, refinement switched off) evolved to $t = 3\Omega^{-1}$ with `rk2` and with `imex2+`, both at `cfl_number = 0.3`. Relative differences of the history columns at $t = 3$:

	mass	E_{tot}	E_{int}	$\langle \rho c_s \rangle$	$\langle \rho w_{\text{grv}} \rangle$	$\langle \rho w_{\text{rey}} \rangle$	$\langle \rho \delta v^2 \rangle$
<code> imex2+ - rk2 / rk2 </code>	2.0×10^{-6}	7.0×10^{-5}	6.3×10^{-5}	3.5×10^{-5}	5.3×10^{-3}	3.2×10^{-4}	3.2×10^{-4}

Two different second-order integrators (three explicit stages of which the first advances by $\gamma\Delta t \approx 1.71\Delta t$, versus two) on a chaotic flow: agreement at 10^{-4} – 10^{-3} in the bulk quantities, with the small gravitational stress — a difference of large terms at this early time — the loosest at 5×10^{-3} (Fig. 2). The spikes of the relative difference of the stress columns at $t \approx 1.75$ and 2.5 in the figure are zero crossings of the stresses themselves (top-left panel), where a relative measure carries no information. This is the level at which `rk2`-validated gravity results transfer to dust runs; the full-size SC14 statistics under `imex2+` are a cluster item and will be attached to the next β -scan submission.

4 The second merge: regression, and the PC2/hybrid couplings

The dust fork’s `dust` branch had moved on by seven commits (to `1f3e1ed9`) while the first merge was based on `d6de1296`; commit `78320aee` merges them (implementation document §4.5). They bring the fork’s own rank-packed deposit exchange (taken in place of ours), CPU particle-mesh optimizations, and two new coupling modes selectable with `<dust>/coupling`: `pc2`, a per-cycle predictor–corrector particle update under `rk2`, and `hybrid`, which switches between PC2 and a first-order split backward-Euler branch on stiffness measures $\zeta = \max \Delta t/t_s$ and $\chi = \max \Delta t \rho_d / (\rho_g t_s)$.

4.1 Regression

Every table of §3 was rerun on the rebuilt serial and MPI binaries. The damping ladder, its three layout variants, the dusty-wave ladder, the C-shock, the drifting-particle test at 1, 2 and 4 ranks, and all multigrid statics (serial and 4 ranks) reproduce the printed digits above exactly; the NSH deviations move within their 10^{-16} round-off (the new non-atomic deposit changes the summation order); and the gas-only gravito-turbulence run is **bit-identical** to the pre-merge history, as it must be — the merge touched nothing on the gas path.

4.2 PC2 coupling under `rk2`

The inputs are the same test inputs with `integrator = rk2` and `coupling = pc2` added to the `<dust>` block (archived under `run/dust4a/merge2/`; the key has to be in the file because the command line cannot add parameters). Damping ladder:

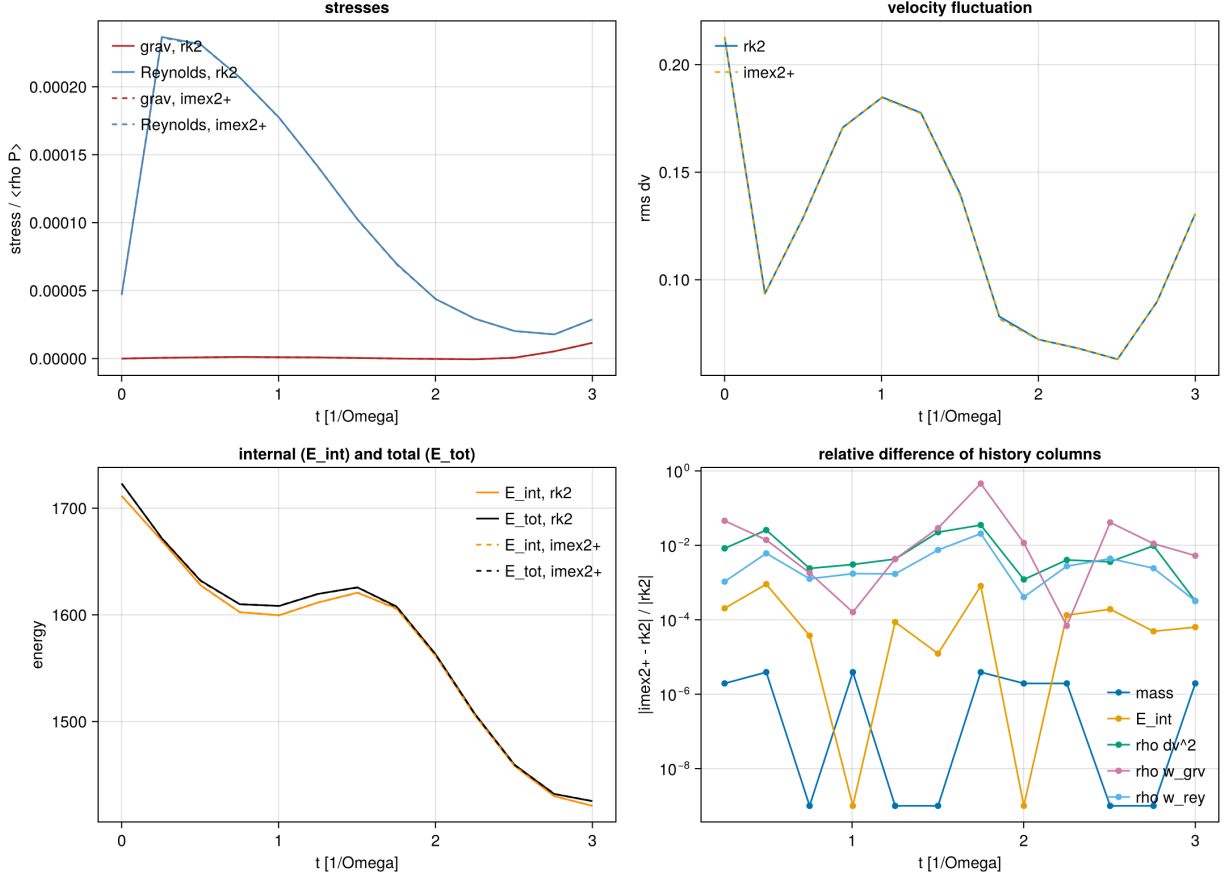


Figure 2: The gas-only gravito-turbulence box under `rk2` (solid) and `imex2+` (dashed): stresses normalized by $\langle \rho P \rangle$, rms velocity fluctuation, internal and total energy, and the relative difference of the history columns.

Δt	err_u	err_{v1}	err_{v2}	err_{v3}	$ \Delta P_{\text{tot}} $
1.235×10^{-2}	2.4349×10^{-7}	2.2596×10^{-7}	2.4748×10^{-7}	1.2039×10^{-6}	2.4×10^{-16}
6.211×10^{-3}	5.2577×10^{-8}	4.8577×10^{-8}	5.3355×10^{-8}	2.5966×10^{-7}	4.5×10^{-17}
3.115×10^{-3}	1.2351×10^{-8}	1.1216×10^{-8}	1.2461×10^{-8}	6.0729×10^{-8}	5.0×10^{-17}
1.558×10^{-3}	2.9941×10^{-9}	2.7081×10^{-9}	3.0167×10^{-9}	1.4707×10^{-8}	5.1×10^{-17}
order of err_u	2.23, 2.10, 2.04	(IMEX ladder: 1.98, 1.99, 1.99)			

Second order, converging onto the Δt^2 slope from above, with errors about 1.4–1.8 $\times$ the IMEX ones at equal Δt and momentum at round-off (Fig. 1, open markers). The dusty wave:

N_x	$L_1(\delta\rho)$	$L_1(\delta v_x)$	order	(IMEX order)
32	1.612×10^{-8}	1.182×10^{-8}	—	—
64	4.483×10^{-9}	3.310×10^{-9}	1.85	1.41
128	1.139×10^{-9}	8.404×10^{-10}	1.98	1.91
256	2.858×10^{-10}	2.109×10^{-10}	2.00	1.95

Cleaner second order than the IMEX ladder and smaller errors at every resolution. The NSH equilibrium under pc2 holds to round-off over the ten orbits (3.2×10^{-16} , 1.3×10^{-16} , 3.1×10^{-16} , 2.1×10^{-16} , 3.6×10^{-16} , 2.4×10^{-16} , 1.0×10^{-17} , 1.7×10^{-16} for the eight components of §3.1). Rank independence with the drifting particles:

ranks	err_u	err_{v1}	err_{v2}	err_{v3}	$ \Delta P_{\text{tot}} $
1	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.5×10^{-14}
2	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	1.9×10^{-15}
4	6.805779×10^{-6}	5.812361×10^{-6}	7.380154×10^{-6}	3.360985×10^{-5}	3.9×10^{-15}

4.3 Hybrid coupling and the split-BE branch

Two runs characterize the fallback. The three-species NSH test under `coupling = hybrid` (automatic selection) reports `pc2_cycles=0 split_be_cycles=4494` with $\zeta = 0.549$: the $t_s = 0.01$ species puts $\Delta t/t_s$ just above the 0.5 entry threshold, so the run stays in the first-order branch throughout, and the equilibrium is no longer an exact fixed point — deviations of $1\text{--}12 \times 10^{-5}$ after ten orbits instead of round-off. This is the designed behavior, not a defect: the first-order splitting error of a stiff fallback, on a problem whose IMEX and PC2 answers are exact. The stiff damping case of the dust tests ($t_s = 10^{-4}, 10^{-3}, 2 \times 10^{-3}$, $\Delta t/t_s$ up to 125) under `hybrid` runs split-BE for all 81 cycles ($\zeta = 32.8$, $\chi = 12.6$) and lands on the exact relaxed state (errors $\leq 4.9 \times 10^{-16}$, momentum 1.1×10^{-16}), the L-stability that the branch exists for. For production runs in the resolved regime the recommendation is therefore `coupling = pc2`; `hybrid` is the safety net for configurations with stopping times below the timestep.

5 Summary

Phase 4a delivers the dust module inside the multigrid tree with both sides certified unchanged: the dust fork’s acceptance numbers and the gravity solver’s documented statics reproduce to every printed digit, serially and at 4 ranks, and the one merge defect (the additive deposit exchange against rank-packed messaging) is fixed with a rank-independence table to show for it. The integrator the dust module’s IMEX coupling imposes, `imex2+`, tracks `rk2` on the stratified gravito-turbulent box at

the level two second-order schemes should — and after the second merge that constraint is optional: the PC2 coupling runs under `rk2` at second order with the same conservation properties, and is the recommended production path in the resolved drag regime.

What 4a does *not* yet provide, and Phase 4b takes up next: the shear-periodic remap of the additive deposits (the module warns that ghost deposits cross the radial faces unremapped, exact only for axisymmetric states), back-reaction with an ideal-gas energy equation (required for cooling), and a particle timestep that does not pay the full-shear-velocity penalty in wide boxes. The tests for those are the 3D shear-periodic NSH state, an azimuthally structured deposit conserved through the remap, and the momentum–energy budget of a drag-coupled ideal gas.